C **Foundations of Programming** Lab Assignments 16 – 22

A#16 – Even/Odd A#17 – Sine Series A#18 – Strings A#19 – Employee A#20 – Point

ASSIGNMENT NO. 21

File Copy using File Handling

Open source.txt for reading and destination.txt for writing. Copy one character at a time using fgetc()/fputc() until EOF.

C CODE

```c
#include <stdio.h>

int main() {
    FILE *src, *dest;
    int  ch;  /* int — NOT char. Required for correct EOF detection */

    src = fopen("source.txt", "r");
    if (src == NULL) {
        printf("Error: Cannot open source.txt\n");
        return 1;
    }

    dest = fopen("destination.txt", "w");
    if (dest == NULL) {
        printf("Error: Cannot create destination.txt\n");
        fclose(src);
        return 1;
    }

    while ((ch = fgetc(src)) != EOF)
        fputc(ch, dest);

    fclose(src);
    fclose(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

SAMPLE I/O

```
source.txt:
Hello
Welcome to C Programming

Output:
File copied successfully.

destination.txt:
Hello
Welcome to C Programming
```

FILE FUNCTIONS

FOPEN(F,"R")
Open for reading

FOPEN(F,"W")
Create/open for writing

FGETC(FP)
Read one char; returns EOF at end

FPUTC(C,FP)
Write one char to file

EOF
End-of-file sentinel = -1

FCLOSE(FP)
Close and flush the file

FLOWCHART

Syntax error in text
mermaid version 10.9.1